



SYDNEY BOYS HIGH
MOORE PARK, SURRY HILLS

2013
HIGHER SCHOOL CERTIFICATE
ASSESSMENT TASK #1

Mathematics Extension 2

General Instructions:

- Reading time – 5 minutes
- Working time – 1.5 hours
- Write using black or blue pen
Black pen is preferred
- Board approved calculators may be used
- A table of standard integrals is provided on the back of the Multiple Choice answer sheet
- All necessary working should be shown in every question if full marks are to be awarded
- Marks may NOT be awarded for messy or badly arranged work
- Answer in simplest exact form unless otherwise stated

Total marks – 70 Marks

Section I Pages 2–3 7 marks

- Attempt Questions 1–7
- Answer on the Multiple Choice answer sheet provided.
- Allow about 10 minutes for this section

Section II Pages 4–6 63 marks

- Attempt Questions 8–10
- Allow about 1 hour 20 minutes for this section
- For Questions 8–10, start a new answer booklet per question

Examiner: Mr D. Hespe

This is an assessment task only and does not necessarily reflect the content or format of the Higher School Certificate.

Section I— 7 marks

Select the alternative A, B, C, or D that best answers the question.

Fill in the response oval on your multiple choice answer sheet.

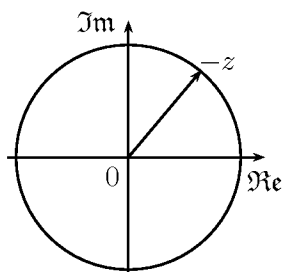
Marks

1. The probability that a randomly chosen angle has a sine which is less than a half is

1

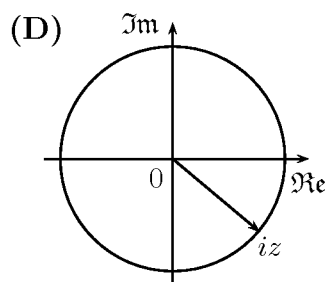
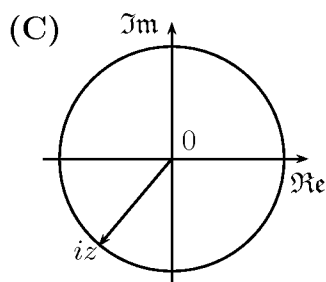
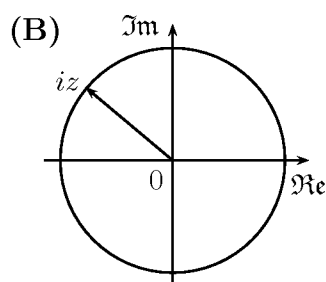
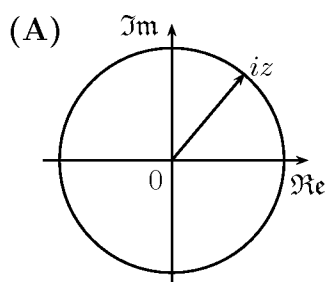
- (A) $\frac{1}{2}$
 (B) $\frac{2\pi}{3}$
 (C) $\frac{2}{3}$
 (D) $\frac{4\pi}{3}$

2.



This graph is a representation of $-z$ shown in an Argand diagram. Which of the following shows iz ?

1



3. If $f(x) = e^{x+2}$ what is the inverse function $f^{-1}(x)$?

1

- (A) $f^{-1}(x) = e^{y-2}$
 (B) $f^{-1}(x) = e^{y+2}$
 (C) $f^{-1}(x) = \log_e x - 2$
 (D) $f^{-1}(x) = \log_e x + 2$

4. What is the domain and range of $y = \cos^{-1}\left(\frac{2x}{5}\right)$? 1

- (A) Domain: $\left[-\frac{5}{2}, \frac{5}{2}\right]$; Range: $[0, \pi]$
- (B) Domain: $[-1, 1]$; Range: $[0, \pi]$
- (C) Domain: $\left[-\frac{5}{2}, \frac{5}{2}\right]$; Range: $[-\pi, \pi]$
- (D) Domain: $[-1, 1]$; Range: $[-\pi, \pi]$

5. Form a polynomial $f(x)$ with real coefficients having the given degree and zeroes.
Degree: 3, Zeroes: $1 + i$ and -5 . 1

- (A) $f(x) = x^3 + x^2 - 8x + 10$
- (B) $f(x) = x^3 - 5x^2 - 8x - 12$
- (C) $f(x) = x^3 + 3x^2 - 8x + 10$
- (D) $f(x) = x^3 + 3x^2 + 10x - 8$

6. Find the indefinite integral $\int 2t^2(1 + t^3)^4 dt$ using the substitution $u = 1 + t^3$. 1

- (A) $\frac{1}{5}(1 + t^3)^5 + c$
- (B) $\frac{2}{5}(1 + t^3)^5 + c$
- (C) $\frac{2}{3}(1 + t^3)^5 + c$
- (D) $\frac{2}{15}(1 + t^3)^5 + c$

7. If $(x - 3)^2 + (y + 2)^2 = 0$, then $x + y =$ 1

- (A) 1
- (B) 2
- (C) 3
- (D) 5

Section II— 63 marks

Marks

Question 8 (21 marks) (use a separate answer booklet)

- (a) You are dealt a hand of 5 cards from a standard deck of 52 (in 4 suits of 13 cards).
- (i) What is the chance that you have a flush (*i.e.* all cards from the same suit)? 1
- (ii) What is the chance of four of a kind (*i.e.* 4 cards of the same value)? 1
- (b) (i) What is $\int \sec^2 \psi \, d\psi$? 1
- (ii) Evaluate $\int_0^\pi \sin^3 \theta \, d\theta$. 3
- (iii) Find $\int \frac{dx}{\sqrt{x^2 - 4x + 20}}$ by using the substitution $y = x - 2$. 3
- (c) Mark on an Argand diagram the points representing the numbers $2 + 3i$ and $-3 + 4i$. 2
- (d) Find, correct to three significant figures, the modulus and argument of $\frac{1}{12 + 5i}$. 2
- (e) Simplify:
- (i) $(5 - 3i)(2 + i)$, 1
- (ii) $\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)$, 1
- (iii) $\frac{1}{5 - 3i} - \frac{1}{5 + 3i}$. 1
- (f) The cubic equation $x^3 + ax - b$ has roots α, β, γ . Given that $\gamma = \alpha\beta$, express each of a and b in terms of γ only, and hence show that $(a + b)^2 = b$. 5

Question 9 (21 marks) (use a separate answer booklet)

- (a) The letters of the word **EXCELLENT** are arranged in a random order. Find the probability that:

(i) the same letter occurs at each end. 2

(ii) **X**, **C** and **N** occur together in any order. 2

(iii) the letters occur in alphabetical order. 1

- (b) Evaluate

(i) $\sin\left(\sin^{-1}\frac{1}{2}\right)$ 1

(ii) $\sin\left(\cos^{-1}\frac{1}{2}\right)$ 1

(iii) $\cos^{-1}\left(\cos\frac{4\pi}{3}\right)$ 1

(iv) Sketch the function $f(x) = -2\pi \cos^{-1}\left(\frac{3x}{\pi}\right)$. 3

(c) Use the substitution $x = \tan \theta$ to evaluate $\int_0^1 \frac{x^2 dx}{(1+x^2)^2}$. 3

(d) (i) Use limits to show why $\frac{dx}{dy} = \frac{1}{\frac{dy}{dx}}$. 1

(ii) Prove that $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$. 3

(iii) Hence find a primitive of $\frac{x^2}{1+x^2}$. 3

Question 10 (21 marks) (use a separate answer booklet)

- (a) A company has to place three orders for supplies among five distributors. Each order is randomly assigned, and a distributor may receive multiple orders. Find the probabilities of the following events.

(i) Each order goes to a different distributor.

1

(ii) All orders go to the same distributor.

1

(iii) Exactly two of the three go to one of the distributors.

1

- (b) (i) Use De Moivre's theorem to express $\cos 3\theta$ in terms of $\cos \theta$ and $\sin 3\theta$ in terms of $\sin \theta$.

2

(ii) Use the result to solve the equation

3

$$8x^3 - 6x + 1 = 0.$$

- (c) When a rational integral function is divided by $(x - \alpha)^2$ the remainder is $R_2(x - \alpha) + R_1$, i.e. $f(x) = (x - \alpha)^2 Q(x) + R_2(x - \alpha) + R_1$.

(i) Prove that $R_1 = f(\alpha)$ and $R_2 = f'(\alpha)$, where $f'(x)$ is the differential coefficient of $f(x)$ with respect to x .

2

(ii) Show that $x^n - nx + n - 1$ is exactly divisible by $(x - 1)^2$ for any integral value of n greater than 1.

2

(iii) Deduce that $2^{4n} - 15n - 1$ is exactly divisible by 225 for any integral value of n greater than 1.

2

- (d) (i) If a, b are the complex numbers represented by points A, B in the Argand diagram, what geometrical properties correspond to the modulus and argument of b/a ?

2

(ii) Show that, if the four points representing the complex numbers z_1, z_2, z_3, z_4 are concyclic, the fraction

5

$$\frac{(z_1 - z_2)(z_3 - z_4)}{(z_3 - z_2)(z_1 - z_4)}$$

must be real.

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Mathematics Extension 2

Solutions

Section I— 7 marks

Select the alternative A, B, C, or D that best answers the question.

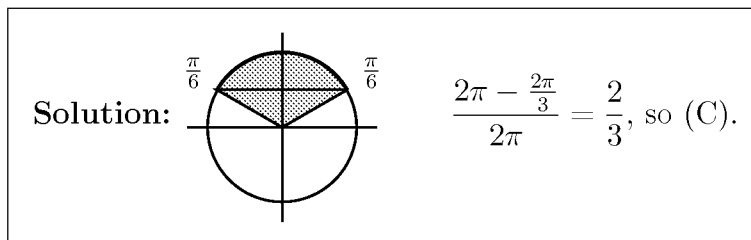
Fill in the response oval on your multiple choice answer sheet.

Marks

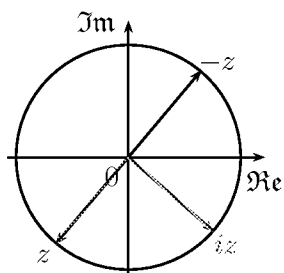
1. The probability that a randomly chosen angle has a sine which is less than a half is

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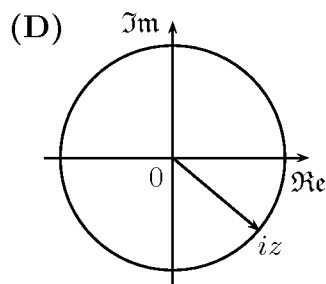
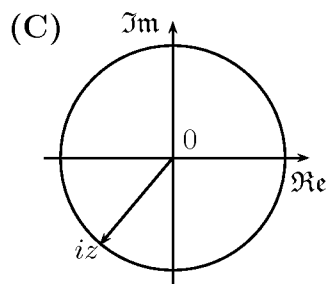
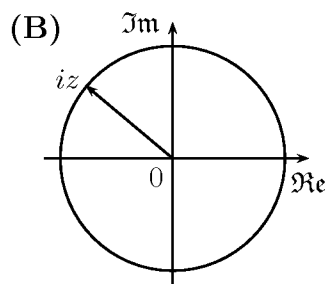
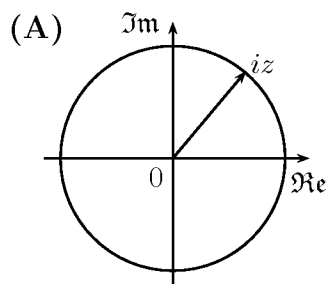
- (A) $\frac{1}{2}$
 (B) $\frac{2\pi}{3}$
 (C) $\frac{2}{3}$
 (D) $\frac{4\pi}{3}$



2.



This graph is a representation of $-z$ shown in an Argand diagram. Which of the following shows iz ?



Solution: z is π from $-z$ and iz is rotated $\frac{\pi}{2}$ anticlockwise from z , so (D).

3. If $f(x) = e^{x+2}$ what is the inverse function $f^{-1}(x)$?

1

- (A) $f^{-1}(x) = e^{y-2}$
 (B) $f^{-1}(x) = e^{y+2}$
 (C) $f^{-1}(x) = \log_e x - 2$
 (D) $f^{-1}(x) = \log_e x + 2$

Solution: $\log_e x = f^{-1}(x) + 2$,
 $f^{-1}(x) = \log_e x - 2$, so (C).

4. What is the domain and range of $y = \cos^{-1}\left(\frac{2x}{5}\right)$? 1

- (A) Domain: $\left[-\frac{5}{2}, \frac{5}{2}\right]$; Range: $[0, \pi]$
 (B) Domain: $[-1, 1]$; Range: $[0, \pi]$
 (C) Domain: $\left[-\frac{5}{2}, \frac{5}{2}\right]$; Range: $[-\pi, \pi]$
 (D) Domain: $[-1, 1]$; Range: $[-\pi, \pi]$

Solution: $-1 \leq \frac{2x}{5} \leq 1,$
 $-\frac{5}{2} \leq x \leq \frac{5}{2}.$
 $0 \leq y \leq \pi,$ so (A).

5. Form a polynomial $f(x)$ with real coefficients having the given degree and zeroes.
 Degree: 3, Zeroes: $1 + i$ and -5 . 1

- (A) $f(x) = x^3 + x^2 - 8x + 10$
 (B) $f(x) = x^3 - 5x^2 - 8x - 12$
 (C) $f(x) = x^3 + 3x^2 - 8x + 10$
 (D) $f(x) = x^3 + 3x^2 + 10x - 8$

Solution: $S_1 = (1 + i) + (1 - i) - 5,$
 $= -3.$
 $S_2 = (1 + i)(1 - i) + (1 + i)(-5) + (1 - i)(-5),$
 $= -8.$
 $S_3 = (1 + i)(1 - i)(-5),$
 $= -10.$

So the solution is $f(x) = x^3 + 3x^2 - 8x + 10$, *i.e.* (C).

6. Find the indefinite integral $\int 2t^2(1 + t^3)^4 dt$ using the substitution $u = 1 + t^3$. 1

- (A) $\frac{1}{5}(1 + t^3)^5 + c$
 (B) $\frac{2}{5}(1 + t^3)^5 + c$
 (C) $\frac{2}{3}(1 + t^3)^5 + c$
 (D) $\frac{2}{15}(1 + t^3)^5 + c$

Solution: $du = 3t^2 dt,$
 $I = \frac{2}{3} \int 3t^2(1 + t^3)^4 dt,$
 $= \frac{2}{3} \int u^4 du,$
 $= \frac{2}{3} \times \frac{u^5}{5} + c,$
 $= \frac{2(1 + t^3)^5}{15},$ so (D).

7. If $(x - 3)^2 + (y + 2)^2 = 0$, then $x + y =$

1

(A) 1

(B) 2

(C) 3

(D) 5

Solution: $(x - 3)^2 \geq 0$, $(y + 2)^2 \geq 0$,
 $(x - 3)^2 + (y + 2)^2 = 0$,
 $\therefore x - 3 = 0$, $y + 2 = 0$,
 $x - 3 + y + 2 = 0$,
 $x + y = 1$, so (A).

Section II— 63 marks

Marks

Question 8 (21 marks) (use a separate answer booklet)

- (a) You are dealt a hand of 5 cards from a standard deck of 52 (in 4 suits of 13 cards).

- (i) What is the chance that you have a flush
(i.e. all cards from the same suit)?

1

Solution: $\frac{4 \times \binom{13}{5}}{\binom{52}{5}} = \frac{33}{16660}.$

- (ii) What is the chance of four of a kind
(i.e. 4 cards of the same value)?

1

Solution: $\frac{13 \times (52 - 4)}{\binom{52}{5}} = \frac{1}{4165}.$

- (b) (i) What is $\int \sec^2 \psi \, d\psi$?

1

Solution: $\tan \psi + c.$

- (ii) Evaluate $\int_0^\pi \sin^3 \theta \, d\theta$.

3

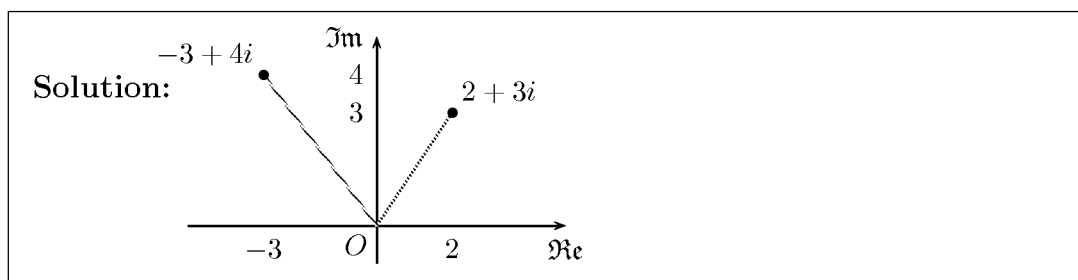
Solution:
$$\begin{aligned} I &= \int_0^\pi \sin^2 \theta \sin \theta \, d\theta, \\ &= \int_0^\pi (1 - \cos^2 \theta) \sin \theta \, d\theta, \\ &= \int_0^\pi \sin \theta \, d\theta - \int_0^\pi \cos^2 \theta \sin \theta \, d\theta, \\ &= [-\cos \theta]_0^\pi - \left[-\frac{\cos^3 \theta}{3} \right]_0^\pi, \\ &= 1 - -1 - \left(\frac{1 - -1}{3} \right), \\ &= \frac{4}{3}. \end{aligned}$$

- (iii) Find $\int \frac{dx}{\sqrt{x^2 - 4x + 20}}$ by using the substitution $y = x - 2$.

3

Solution: $x^2 - 4x + 4 + 16 = (x - 2)^2 + 4^2$,
 $dy = dx$,
 $I = \int \frac{dy}{\sqrt{y^2 + 4^2}}$,
 $= \ln(y + \sqrt{y^2 + 4^2}) + c$,
 $= \ln(x - 2 + \sqrt{x^2 - 4x + 20}) + c$.

- (c) Mark on an Argand diagram the points representing the numbers $2 + 3i$ and $-3 + 4i$. 2



- (d) Find, correct to three significant figures, the modulus and argument of $\frac{1}{12 + 5i}$. 2

Solution: $|12 + 5i| = \sqrt{144 + 25}$,
 $= 13$.
 $\arg(12 + 5i) = \tan^{-1}\left(\frac{5}{12}\right)$,
 ≈ 0.395 or 22.6° .
 So $\left|\frac{1}{12 + 5i}\right| = \frac{1}{13}$,
 ≈ 0.0769 .
 $\arg\left(\frac{1}{12 + 5i}\right) \approx -0.395$.

- (e) Simplify:

(i) $(5 - 3i)(2 + i)$, 1

Solution: $10 + 5i - 6i + 3 = 13 - i$.

(ii) $\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)$, 1

Solution: $\cos \frac{\pi}{2} + i \sin \frac{\pi}{2} = i$.

(iii) $\frac{1}{5 - 3i} - \frac{1}{5 + 3i}$. 1

Solution: $\frac{5 + 3i - (5 - 3i)}{25 + 9} = \frac{3i}{17}$.

- (f) The cubic equation $x^3 + ax - b$ has roots α, β, γ . Given that $\gamma = \alpha\beta$, express each of a and b in terms of γ only, and hence show that $(a + b)^2 = b$.

5

Solution:

$$\begin{aligned}0 &= \alpha + \beta + \gamma, \\ \alpha + \beta &= -\gamma, \\ a &= \alpha\beta + \beta\gamma + \gamma\alpha, \\ &= \gamma + \gamma(\alpha + \beta), \\ &= \gamma - \gamma^2. \\ b &= \alpha\beta\gamma, \\ &= \gamma^2. \\ a + b &= \gamma, \\ (a + b)^2 &= \gamma^2, \\ &= b.\end{aligned}$$

Question 9 (21 marks) (use a separate answer booklet)

- (a) The letters of the word **EXCELLENT** are arranged in a random order. Find the probability that:

- (i) the same letter occurs at each end.

2

$$\text{Solution: } \frac{\frac{7!}{2!} + \frac{7!}{3!}}{\frac{9!}{3! \times 2!}} = \frac{1}{9}.$$

- (ii) **X**, **C** and **N** occur together in any order.

2

$$\text{Solution: } \frac{\frac{3! \times 7!}{3! \times 2!}}{\frac{9!}{3! \times 2!}} = \frac{1}{12}.$$

- (iii) the letters occur in alphabetical order.

1

$$\text{Solution: } \frac{1}{\frac{9!}{3! \times 2!}} = \frac{1}{30240}$$

- (b) Evaluate

(i) $\sin\left(\sin^{-1}\frac{1}{2}\right)$

1

$$\text{Solution: } \frac{1}{2}$$

(ii) $\sin\left(\cos^{-1}\frac{1}{2}\right)$

1

$$\text{Solution: } \sin\frac{\pi}{3} = \frac{\sqrt{3}}{2}$$

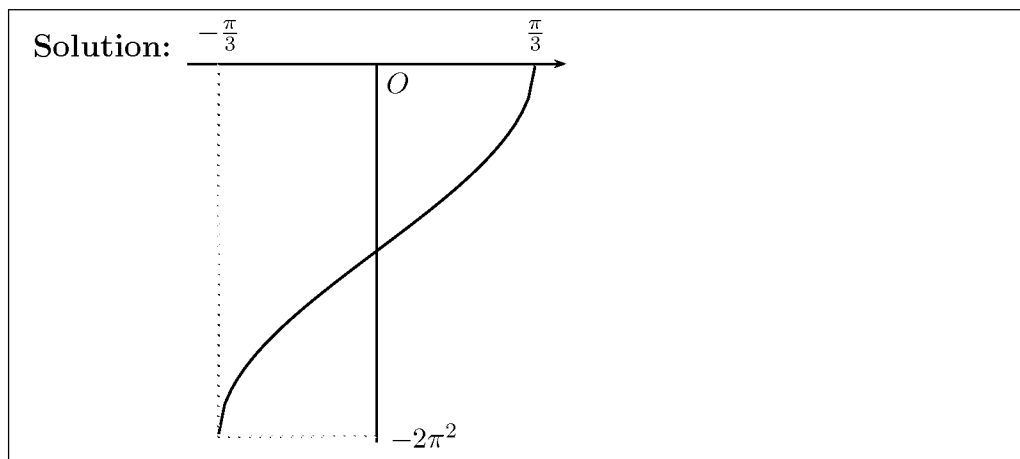
(iii) $\cos^{-1}\left(\cos\frac{4\pi}{3}\right)$

1

$$\text{Solution: } \cos^{-1}\left(-\frac{1}{2}\right) = \frac{2\pi}{3}$$

- (iv) Sketch the function $f(x) = -2\pi \cos^{-1}\left(\frac{3x}{\pi}\right)$.

3



- (c) Use the substitution $x = \tan \theta$ to evaluate $\int_0^1 \frac{x^2 dx}{(1+x^2)^2}$.

3

Solution:

$$\begin{aligned}
 dx &= \sec^2 \theta d\theta, \\
 x = 0 &\implies \theta = 0, \\
 x = 1 &\implies \theta = \frac{\pi}{4}, \\
 I &= \int_0^{\frac{\pi}{4}} \frac{\tan^2 \theta \sec^2 \theta d\theta}{(1 + \tan^2 \theta)^2}, \\
 &= \int_0^{\frac{\pi}{4}} \frac{\sin^2 \theta}{\cos^2 \theta} \times \frac{d\theta}{\cos^2 \theta} \times \frac{1}{(\sec^2 \theta)^2}, \\
 &= \int_0^{\frac{\pi}{4}} \sin^2 \theta d\theta, \\
 &= \int_0^{\frac{\pi}{4}} \frac{1 - \cos 2\theta}{2} d\theta, \\
 &= \frac{1}{2} \left[\theta - \frac{\sin 2\theta}{2} \right]_0^{\frac{\pi}{4}}, \\
 &= \frac{1}{2} \left(\frac{\pi}{4} - \frac{1}{2} - (0 - 0) \right), \\
 &= \frac{\pi - 2}{8}.
 \end{aligned}$$

- (d) (i) Use limits to show why $\frac{dx}{dy} = \frac{1}{\frac{dy}{dx}}$.

1

Solution: We consider a small change in x , δx and a corresponding small change in y , δy .

$$\begin{aligned}
 \text{Now } \frac{dx}{dy} &= \lim_{\delta y \rightarrow 0} \frac{\delta x}{\delta y}, \\
 &= \lim_{\delta y \rightarrow 0} \frac{1}{\frac{\delta y}{\delta x}}
 \end{aligned}$$

If y is a continuous function of x then $\delta y \rightarrow 0$ as $\delta x \rightarrow 0$;

$$\begin{aligned}\text{whence } \frac{dx}{dy} &= \lim_{\delta x \rightarrow 0} \frac{1}{\frac{\delta y}{\delta x}}, \\ &= \frac{1}{\frac{dy}{dx}}.\end{aligned}$$

(ii) Prove that $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$.

3

Solution: Let $y = \tan^{-1} x$, $x \in \mathbb{R}$, $-\frac{\pi}{2} < y < \frac{\pi}{2}$,

$$\begin{aligned}x &= \tan y, \\ \frac{dx}{dy} &= \sec^2 y, \\ &= 1 + \tan^2 y, \\ &= 1 + x^2. \\ \therefore \frac{dy}{dx} &= \frac{1}{1+x^2}, \quad x \in \mathbb{R}, \quad \frac{dy}{dx} > 0.\end{aligned}$$

(iii) Hence find a primitive of $\frac{x^2}{1+x^2}$.

3

Solution:
$$\begin{aligned}\int \frac{x^2 dx}{1+x^2} &= \int \left(1 - \frac{1}{1+x^2}\right) dx, \\ &= x - \tan^{-1} x + c.\end{aligned}$$

Question 10 (21 marks) (use a separate answer booklet)

- (a) A company has to place three orders for supplies among five distributors. Each order is randomly assigned, and a distributor may receive multiple orders. Find the probabilities of the following events.

- (i) Each order goes to a different distributor.

1

Solution: Total of all arrangements = $5 \times 5 \times 5$,
 $= 125$.
 Ways of distributing = $5 \times 4 \times 3$,
 $= 60$.
 $\therefore P(\text{each different}) = \frac{60}{125}$,
 $= \frac{12}{25}$.

- (ii) All orders go to the same distributor.

1

Solution: There are only 5 ways this can occur,
 so $P(\text{all to same}) = \frac{5}{125}$,
 $= \frac{1}{25}$.

- (iii) Exactly two of the three go to one of the distributors.

1

Solution: Method 1—

This is the only other possibility
 so $P(2 \text{ of } 3 \text{ to } 1) = 1 - \left(\frac{12}{25} + \frac{1}{25}\right)$,
 $= \frac{12}{25}$.

Method 2—

$\binom{3}{2} \times 5 \times 4 = 60$.
 So $P(2 \text{ of } 3 \text{ to } 1) = \frac{60}{125}$,
 $= \frac{12}{25}$.

- (b) (i) Use De Moivre's theorem to express $\cos 3\theta$ in terms of $\cos \theta$ and $\sin 3\theta$ in terms of $\sin \theta$.

2

Solution: $\cos 3\theta + i \sin 3\theta = (\cos \theta + i \sin \theta)^3$,
 $= \cos^3 \theta + 3i \cos^2 \theta \sin \theta - 3 \cos \theta \sin^2 \theta - i \sin^3 \theta$.
 $\cos 3\theta = \cos^3 \theta - 3 \cos \theta \sin^2 \theta$ (real part),
 $= \cos^3 \theta - 3 \cos \theta (1 - \cos^2 \theta)$,
 $= 4 \cos^3 \theta - 3 \cos \theta$.
 $\sin 3\theta = 3 \cos^2 \theta \sin \theta - \sin^3 \theta$ (imaginary part),
 $= 3(1 - \sin^2 \theta) \sin \theta - \sin^3 \theta$,
 $= 3 \sin \theta - 4 \sin^3 \theta$.

- (ii) Use the result to solve the equation

$$8x^3 - 6x + 1 = 0.$$

Solution: If we put $x = \cos \theta$, then

$$8 \cos^3 \theta - 6 \cos \theta + 1 = 0,$$

$$4 \cos^3 \theta - 3 \cos \theta = -\frac{1}{2},$$

$$\text{i.e. } \cos 3\theta = -\frac{1}{2},$$

$$3\theta = \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{8\pi}{3}, \frac{10\pi}{3} \dots$$

$$\theta = \frac{2\pi}{9}, \frac{4\pi}{9}, \frac{8\pi}{9}, \frac{10\pi}{9} \dots$$

$$x = \cos \frac{2\pi}{9}, \cos \frac{4\pi}{9}, \cos \frac{8\pi}{9}, \cos \frac{10\pi}{9} \dots$$

As it is a cubic, only 3 values of x are possible.

$$\text{Now, } \cos \frac{8\pi}{9} = -\cos \frac{\pi}{9}, \cos \frac{10\pi}{9} = -\cos \frac{\pi}{9}, \text{ etc.}$$

$$\text{Hence } x = -\cos \frac{\pi}{9}, \cos \frac{2\pi}{9}, \cos \frac{4\pi}{9} \text{ only.}$$

- (c) When a rational integral function is divided by $(x - \alpha)^2$ the remainder is $R_2(x - \alpha) + R_1$, i.e. $f(x) = (x - \alpha)^2 Q(x) + R_2(x - \alpha) + R_1$.

- (i) Prove that $R_1 = f(\alpha)$ and $R_2 = f'(\alpha)$, where $f'(x)$ is the differential coefficient of $f(x)$ with respect to x .

Solution: $f(x) = (x - \alpha)^2 Q(x) + R_2(x - \alpha) + R_1 \dots \dots \dots \boxed{1}$

$$f(\alpha) = (\alpha - \alpha)^2 Q(\alpha) + R_2(\alpha - \alpha) + R_1,$$

$$= 0 + 0 + R_1.$$

$$\text{i.e. } R_1 = f(\alpha).$$

From $\boxed{1}$, differentiating w.r.t. x ,

$$f'(x) = 2(x - \alpha)Q(x) + (x - \alpha)^2 Q'(x) + R_2 + (x - \alpha)R_2',$$

$$\therefore f'(\alpha) = 2(\alpha - \alpha)Q(\alpha) + (\alpha - \alpha)^2 Q'(\alpha) + R_2 + (\alpha - \alpha)R_2',$$

$$= 0 + 0 + R_2 + 0,$$

$$\text{i.e. } R_2 = f'(\alpha).$$

- (ii) Show that $x^n - nx + n - 1$ is exactly divisible by $(x - 1)^2$ for any integral value of n greater than 1.

Solution: Let $f(x) = x^n - nx + n - 1$,

$$f(1) = 1^n - n + n - 1,$$

$$= 0 = R_1.$$

$$f'(x) = nx^{n-1} - n,$$

$$f'(1) = n.1^{n-1} - n,$$

$$= n - n = 0 = R_2.$$

$$\therefore f(x) = (x - 1)^2 Q(x) + 0 + 0,$$

$$\text{i.e. } x^n - nx + n - 1 \text{ is divisible by } (x - 1)^2.$$

n must be an integer since $x^n - nx + n - 1$ must be a rational function before the above procedure can be used.

- (iii) Deduce that $2^{4n} - 15n - 1$ is exactly divisible by 225 for any integral value of n greater than 1. 2

Solution: $x^n - nx + n - 1 = x^n - (x - 1)n - 1,$

Taking the $x = 16$ we have

$$\begin{aligned} x^n - (x - 1)n - 1 &= 16^n - 15n - 1, \\ &= 2^{4n} - 15n - 1. \end{aligned}$$

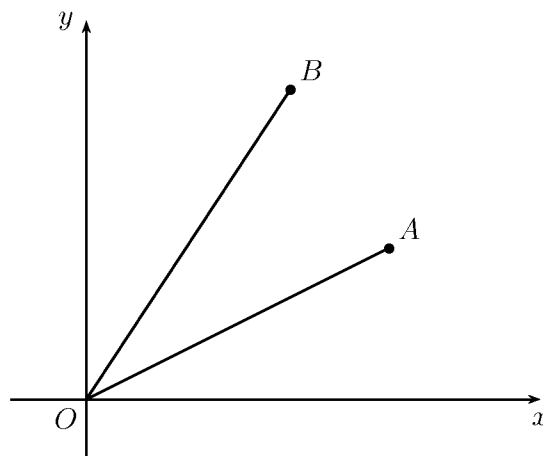
However $x^n - (x - 1)n - 1$ is divisible by $(x - 1)^2$, $n > 1$.

$\therefore 2^{4n} - 15n - 1$ is divisible by $(15)^2$ for any integral value of $n > 1$.

i.e. $2^{4n} - 15n - 1$ is divisible by 225 for any integral value of n greater than 1.

- (d) (i) If a, b are the complex numbers represented by points A, B in the Argand diagram, what geometrical properties correspond to the modulus and argument of b/a ? 2

Solution:



The lengths of the vectors $\vec{OA}$, $\vec{OB}$ are equal respectively to $|a|$ and $|b|$ and the angles xOA , xOB are equal to $\arg a$ and $\arg b$.

Since $\left| \frac{b}{a} \right| = \frac{|b|}{|a|} = \frac{OB}{OA},$

the modulus of b/a corresponds to the ratio OB/OA .

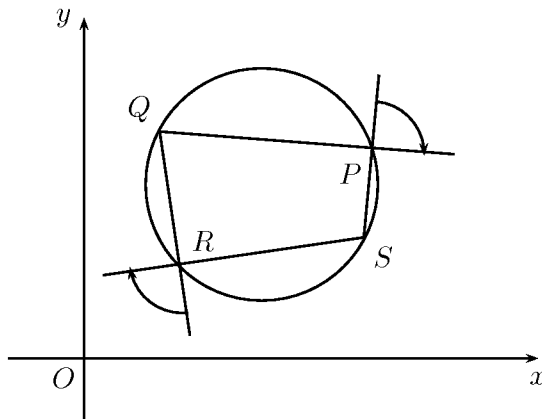
Again, since $\arg \left(\frac{b}{a} \right) = \arg b - \arg a = \angle xOB - \angle xOA,$
 $= \angle AOB.$

- (ii) Show that, if the four points representing the complex numbers z_1, z_2, z_3, z_4 are concyclic, the fraction 5

$$\frac{(z_1 - z_2)(z_3 - z_4)}{(z_3 - z_2)(z_1 - z_4)}$$

must be real.

Solution: Let the representative points of the four complex numbers z_1, z_2, z_3, z_4 be P, Q, R, S .



Clearly, the sum of the marked angles is π (opposite $\angle$ s of a cyclic quad. and vert. opp. $\angle$ s). Since vectors $\vec{SP}$, $\vec{QP}$ represent respectively the complex numbers $z_4 - z_1$ and $z_2 - z_1$, the angle of turn from $\vec{SP}$ to $\vec{QP}$ is

$$\arg \left(\frac{z_2 - z_1}{z_4 - z_1} \right).$$

Similarly the angle of turn from $\vec{QR}$ to $\vec{SR}$ is $\arg \left(\frac{z_4 - z_3}{z_2 - z_3} \right)$. Hence

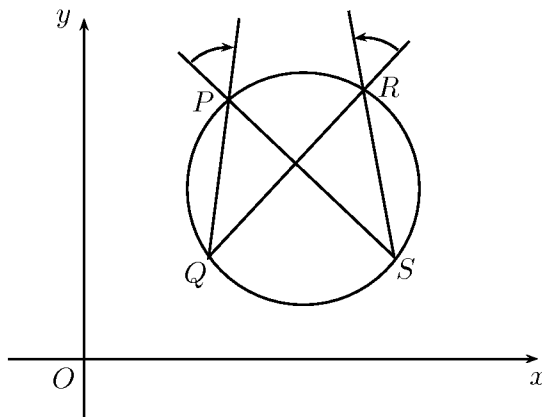
$$\arg \left(\frac{z_2 - z_1}{z_4 - z_1} \right) + \arg \left(\frac{z_4 - z_3}{z_2 - z_3} \right) = \pi.$$

But the sum of the arguments of two complex numbers is equal to the argument of their product, so that

$$\arg \left(\frac{z_2 - z_1}{z_4 - z_1} \times \frac{z_4 - z_3}{z_2 - z_3} \right) = \pi.$$

If the argument of a complex number is π , the number is a negative real number as required.

NOTE: even if we label the points around the circle in some other, non-cyclic order:



Then the sum of the arguments is zero and we have a positive real number.

End of Paper

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